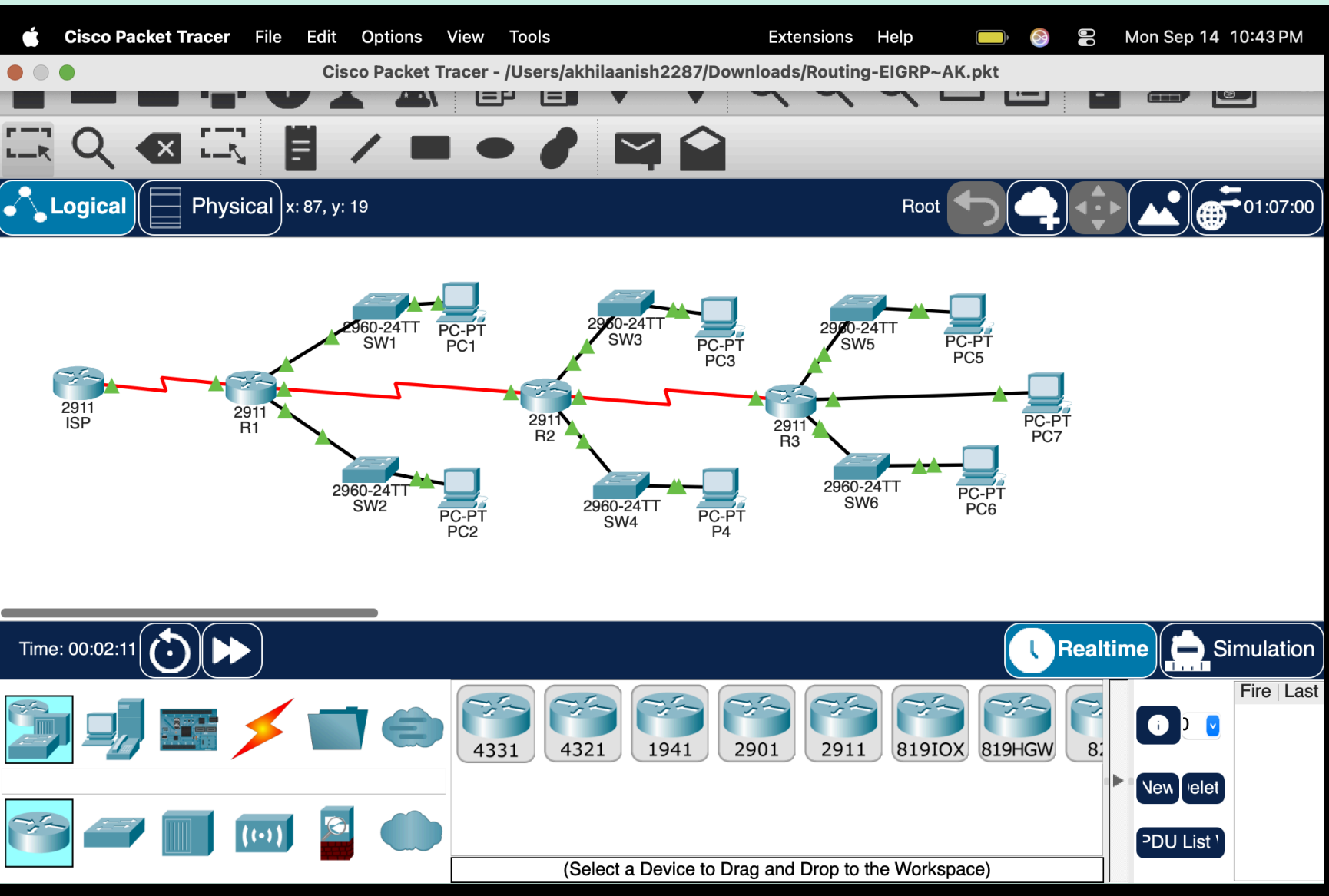




TOPOLOGY OF THE GIVEN ASSIGNMENT 🖱️

```
C:\>ping 192.168.10.1
```

```
Pinging 192.168.10.1 with 32 bytes of data:
```

```
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
```

```
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
```

```
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
```

```
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 192.168.10.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>ping 192.168.20.10
```

```
Pinging 192.168.20.10 with 32 bytes of data:
```

```
Reply from 192.168.10.1: Destination host unreachable.
```

```
Request timed out.
```

```
Reply from 192.168.10.1: Destination host unreachable.
```

```
Reply from 192.168.10.1: Destination host unreachable.
```

```
Ping statistics for 192.168.20.10:
```

```
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>ping 192.168.30.10
```

```
Pinging 192.168.30.10 with 32 bytes of data:
```

```
Reply from 192.168.10.1: Destination host unreachable.
```

```
Reply from 192.168.10.1: Destination host unreachable.
```

```
Reply from 192.168.10.1: Destination host unreachable.
```

```
Request timed out.
```

```
Ping statistics for 192.168.30.10:
```

```
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>akhilaaal
```

Logical

Time: 02:21:19

R3

Physical Config CLI Attributes

IOS Command Line Interface

	0	10.0.23.1	Se0/0/0	(sec)	(ms)	Cnt	Num
				14	00:48:43	40	1000 0 12

```

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
D       10.0.12.0/30 [90/2681856] via 10.0.23.1, 00:54:22, Serial0/0/0
C       10.0.23.0/30 is directly connected, Serial0/0/0
L       10.0.23.2/32 is directly connected, Serial0/0/0
D       192.168.10.0/24 [90/2684416] via 10.0.23.1, 00:54:22, Serial0/0/0
D       192.168.11.0/24 [90/2684416] via 10.0.23.1, 00:54:22, Serial0/0/0
D       192.168.20.0/24 [90/2172416] via 10.0.23.1, 00:54:22, Serial0/0/0
D       192.168.21.0/24 [90/2172416] via 10.0.23.1, 00:54:22, Serial0/0/0
       192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0
       192.168.31.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.31.0/24 is directly connected, GigabitEthernet0/1
L       192.168.31.1/32 is directly connected, GigabitEthernet0/1

Router#
Router#
Router#
Router#akhilaaa
                    
```

Copy

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☐ Top

R2

Physical Config CLI Attributes

IOS Command Line Interface

Router#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

* - candidate default, U - per-user static route, o - ODR

P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks

C 10.0.12.0/30 is directly connected, Serial0/0/0

L 10.0.12.2/32 is directly connected, Serial0/0/0

C 10.0.23.0/30 is directly connected, Serial0/0/1

L 10.0.23.1/32 is directly connected, Serial0/0/1

D 192.168.10.0/24 [90/2172416] via 10.0.12.1, 00:56:31, Serial0/0/0

D 192.168.11.0/24 [90/2172416] via 10.0.12.1, 00:56:31, Serial0/0/0

192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.20.0/24 is directly connected, GigabitEthernet0/0

L 192.168.20.1/32 is directly connected, GigabitEthernet0/0

192.168.21.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.21.0/24 is directly connected, GigabitEthernet0/1

L 192.168.21.1/32 is directly connected, GigabitEthernet0/1

D 192.168.30.0/24 [90/2172416] via 10.0.23.2, 00:53:54, Serial0/0/1

D 192.168.31.0/24 [90/2172416] via 10.0.23.2, 00:53:54, Serial0/0/1

Router#

Router#

Router#

Router#

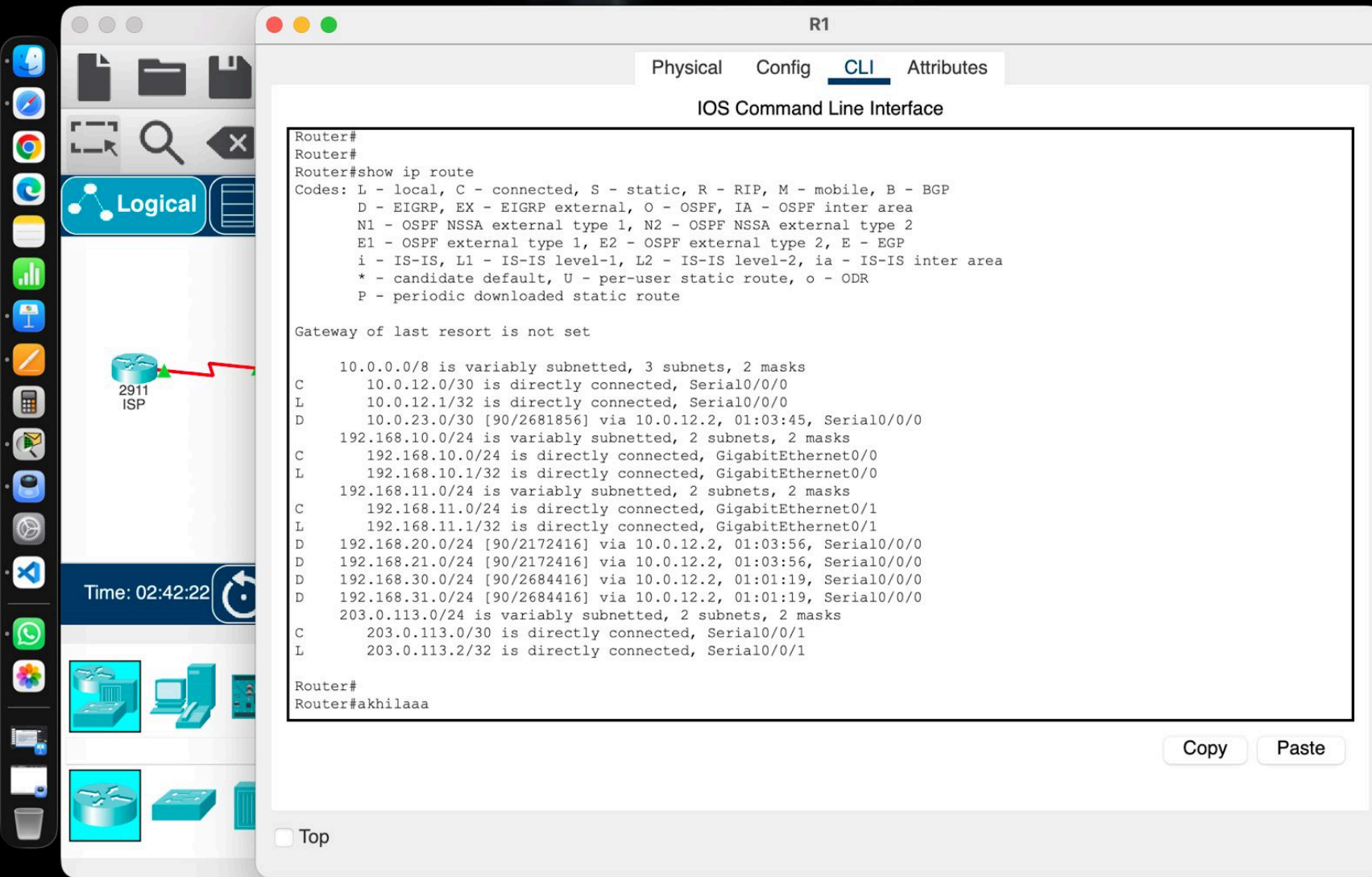
Router#

Router#akhilaaa

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☐ Top



Logical

Time: 02:42:22

2911 ISP

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.12.0/30 is directly connected, Serial0/0/0
L       10.0.12.1/32 is directly connected, Serial0/0/0
D       10.0.23.0/30 [90/2681856] via 10.0.12.2, 01:03:45, Serial0/0/0
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
    192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/24 is directly connected, GigabitEthernet0/1
L       192.168.11.1/32 is directly connected, GigabitEthernet0/1
D       192.168.20.0/24 [90/2172416] via 10.0.12.2, 01:03:56, Serial0/0/0
D       192.168.21.0/24 [90/2172416] via 10.0.12.2, 01:03:56, Serial0/0/0
D       192.168.30.0/24 [90/2684416] via 10.0.12.2, 01:01:19, Serial0/0/0
D       192.168.31.0/24 [90/2684416] via 10.0.12.2, 01:01:19, Serial0/0/0
    203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, Serial0/0/1
L       203.0.113.2/32 is directly connected, Serial0/0/1

Router#
Router#akhilaaa
```

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☐ Top

R3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>
Router>
Router>enable
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.23.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
D    10.0.12.0/30 [90/2681856] via 10.0.23.1, 01:30:25, Serial0/0/0
C    10.0.23.0/30 is directly connected, Serial0/0/0
L    10.0.23.2/32 is directly connected, Serial0/0/0
D    192.168.10.0/24 [90/2684416] via 10.0.23.1, 01:30:25, Serial0/0/0
D    192.168.11.0/24 [90/2684416] via 10.0.23.1, 01:30:25, Serial0/0/0
D    192.168.20.0/24 [90/2172416] via 10.0.23.1, 01:30:25, Serial0/0/0
D    192.168.21.0/24 [90/2172416] via 10.0.23.1, 01:30:25, Serial0/0/0
    192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, GigabitEthernet0/0
L    192.168.30.1/32 is directly connected, GigabitEthernet0/0
    192.168.31.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.31.0/24 is directly connected, GigabitEthernet0/1
L    192.168.31.1/32 is directly connected, GigabitEthernet0/1
D*EX 0.0.0.0/0 [170/2707456] via 10.0.23.1, 00:01:35, Serial0/0/0

Router#akhilaaaa
```

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R2

Physical Config CLI Attributes

IOS Command Line Interface

```
L    192.168.21.1/32 is directly connected, GigabitEthernet0/1
D    192.168.30.0/24 [90/2172416] via 10.0.23.2, 01:27:39, Serial0/0/1
D    192.168.31.0/24 [90/2172416] via 10.0.23.2, 01:27:39, Serial0/0/1

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.12.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.0.12.0/30 is directly connected, Serial0/0/0
L    10.0.12.2/32 is directly connected, Serial0/0/0
C    10.0.23.0/30 is directly connected, Serial0/0/1
L    10.0.23.1/32 is directly connected, Serial0/0/1
D    192.168.10.0/24 [90/2172416] via 10.0.12.1, 01:32:04, Serial0/0/0
D    192.168.11.0/24 [90/2172416] via 10.0.12.1, 01:32:04, Serial0/0/0
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/0
L    192.168.20.1/32 is directly connected, GigabitEthernet0/0
    192.168.21.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.21.0/24 is directly connected, GigabitEthernet0/1
L    192.168.21.1/32 is directly connected, GigabitEthernet0/1
D    192.168.30.0/24 [90/2172416] via 10.0.23.2, 01:29:27, Serial0/0/1
D    192.168.31.0/24 [90/2172416] via 10.0.23.2, 01:29:27, Serial0/0/1
D*EX 0.0.0.0/0 [170/2195456] via 10.0.12.1, 00:00:37, Serial0/0/0

Router#akhilaaa
```

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Logical

Time: 02:54:51

2911 ISP

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
S* 0.0.0.0/0 [1/0] via 203.0.113.1

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.12.0/30 is directly connected, Serial0/0/0
L       10.0.12.1/32 is directly connected, Serial0/0/0
D       10.0.23.0/30 [90/2681856] via 10.0.12.2, 01:31:02, Serial0/0/0
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
    192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.11.0/24 is directly connected, GigabitEthernet0/1
L       192.168.11.1/32 is directly connected, GigabitEthernet0/1
D       192.168.20.0/24 [90/2172416] via 10.0.12.2, 01:31:13, Serial0/0/0
D       192.168.21.0/24 [90/2172416] via 10.0.12.2, 01:31:13, Serial0/0/0
D       192.168.30.0/24 [90/2684416] via 10.0.12.2, 01:28:36, Serial0/0/0
D       192.168.31.0/24 [90/2684416] via 10.0.12.2, 01:28:36, Serial0/0/0
    203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, Serial0/0/1
L       203.0.113.2/32 is directly connected, Serial0/0/1
S* 0.0.0.0/0 [1/0] via 203.0.113.1

Router#configure terminal
```

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R1

Router#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C 10.0.12.0/30 is directly connected, Serial0/0/0
L 10.0.12.1/32 is directly connected, Serial0/0/0
D 10.0.23.0/30 [90/2681856] via 10.0.12.2, 02:36:40, Serial0/0/0
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.10.0/24 is directly connected, GigabitEthernet0/0
L 192.168.10.1/32 is directly connected, GigabitEthernet0/0
192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.11.0/24 is directly connected, GigabitEthernet0/1
L 192.168.11.1/32 is directly connected, GigabitEthernet0/1
D 192.168.20.0/24 [90/2172416] via 10.0.12.2, 02:36:51, Serial0/0/0
D 192.168.21.0/24 [90/2172416] via 10.0.12.2, 02:36:51, Serial0/0/0
D 192.168.30.0/24 [90/2684416] via 10.0.12.2, 02:34:14, Serial0/0/0
D 192.168.31.0/24 [90/2684416] via 10.0.12.2, 02:34:14, Serial0/0/0
203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C 203.0.113.0/30 is directly connected, Serial0/0/1
L 203.0.113.2/32 is directly connected, Serial0/0/1
S* 0.0.0.0/0 [1/0] via 203.0.113.1

Router#

Router#

Router#ping 203.0.113.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:

!!!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 8/12/23 ms

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Router#show ip protocols

Routing Protocol is "eigrp 100 "

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Default networks flagged in outgoing updates

Default networks accepted from incoming updates

Redistributing: eigrp 100, static

EIGRP-IPv4 Protocol for AS(100)

Metric weight K1=1, K2=0, K3=1, K4=0, K5=0

NSF-aware route hold timer is 240

Router-ID: 10.0.12.1

Topology : 0 (base)

Active Timer: 3 min

Distance: internal 90 external 170

Maximum path: 4

Maximum hopcount 100

Maximum metric variance 1

Automatic Summarization: disabled

Automatic address summarization:

Maximum path: 4

Routing for Networks:

192.168.10.0

192.168.11.0

10.0.12.0/30

Routing Information Sources:

Gateway	Distance	Last Update
---------	----------	-------------

10.0.12.2	90	2743713
-----------	----	---------

Distance: internal 90 external 170

Router#

Router#show ip eigrp neighbors

IP-EIGRP neighbors for process 100

H	Address	Interface	Hold Uptime (sec)	SRTT (ms)	RTO	Q Cnt	Seq Num
0	10.0.12.2	Se0/0/0	11 02:36:45	40	1000	0	14

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Logical

Time: 04:21:37

2911 ISP

R3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.30.1    YES manual up          up
GigabitEthernet0/1       192.168.31.1    YES manual up          up
GigabitEthernet0/2       192.168.99.1    YES manual up          up
Serial0/0/0              10.0.23.2       YES manual up          up
Serial0/0/1              unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down

Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.0.23.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
    D    10.0.12.0/30 [90/2681856] via 10.0.23.1, 02:58:47, Serial0/0/0
    C    10.0.23.0/30 is directly connected, Serial0/0/0
    L    10.0.23.2/32 is directly connected, Serial0/0/0
    D    192.168.10.0/24 [90/2684416] via 10.0.23.1, 02:58:47, Serial0/0/0
    D    192.168.11.0/24 [90/2684416] via 10.0.23.1, 02:58:47, Serial0/0/0
    D    192.168.20.0/24 [90/2172416] via 10.0.23.1, 02:58:47, Serial0/0/0
    D    192.168.21.0/24 [90/2172416] via 10.0.23.1, 02:58:47, Serial0/0/0
    192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
    C    192.168.30.0/24 is directly connected, GigabitEthernet0/0
    L    192.168.30.1/32 is directly connected, GigabitEthernet0/0
    192.168.31.0/24 is variably subnetted, 2 subnets, 2 masks
    C    192.168.31.0/24 is directly connected, GigabitEthernet0/1
```

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R1 Routing Table (show ip route)

R1# show ip route

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile,
B - BGP, D - EIGRP, EX - EIGRP external
O - OSPF, IA - OSPF inter area, N1 - OSPF NSSA external type 1,
N2 - OSPF NSSA external type 2, E1 - OSPF external type 1,
E2 - OSPF external type 2, E - EGP, i - IS-IS, l1 - IS-IS level-1,
l2 - IS-IS level-2, * - candidate default, U - per-user static route,
o - ODR, P - periodic downloaded static route

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

```
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0
C    192.168.11.0/24 is directly connected, GigabitEthernet0/0
D    192.168.20.0/24 [90/2684416] via 10.0.12.2, 00:01:23, Serial0/0/0
D    192.168.21.0/24 [90/2684416] via 10.0.12.2, 00:01:23, Serial0/0/0
D    192.168.30.0/24 [90/2684416] via 10.0.12.2, 00:01:23, Serial0/0/0
D    192.168.31.0/24 [90/2684416] via 10.0.12.2, 00:01:23, Serial0/0/0
D    192.168.99.0/24 [90/2684416] via 10.0.12.2, 00:01:23, Serial0/0/0
S*   0.0.0.0/0 [1/0] via 203.0.113.1
```

R1#

R3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet0/0
Router(config-if)#shutdown

Router(config-if)#end
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to down

Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface                IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0       192.168.30.1    YES manual administratively down  down
GigabitEthernet0/1       192.168.31.1    YES manual up                    up
GigabitEthernet0/2       192.168.99.1    YES manual up                    up
Serial0/0/0              10.0.23.2       YES manual up                    up
Serial0/0/1              unassigned      YES unset  administratively down  down
Vlan1                    unassigned      YES unset  administratively down  down
Router#show ip protocols

Routing Protocol is "eigrp 100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
    EIGRP-IPv4 Protocol for AS(100)
      Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
      NSF-aware route hold timer is 240
```

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R3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#show ip eigrp 100
EIGRP-IPv4 Protocol for AS(100)
  Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
  NSF-aware route hold timer is 240
  Router-ID: 10.0.23.2
  Topology : 0 (base)
    Active Timer: 3 min
    Distance: internal 90 external 170
    Maximum path: 4
    Maximum hopcount 100
    Maximum metric variance 1

  Automatic Summarization: disabled
  Automatic address summarization:
  Maximum path: 4
  Routing for Networks:
    192.168.30.0
    192.168.31.0
    10.0.23.0/30
    192.168.99.0
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.23.1        90            2186597
  Distance: internal 90 external 170

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router eigrp 100
Router(config-router)#no network 192.168.31.0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

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R3

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#
Router#show ip protocols

Routing Protocol is "eigrp 100 "
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  Redistributing: eigrp 100
    EIGRP-IPv4 Protocol for AS(100)
      Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
      NSF-aware route hold timer is 240
      Router-ID: 10.0.23.2
      Topology : 0 (base)
        Active Timer: 3 min
        Distance: internal 90 external 170
        Maximum path: 4
        Maximum hopcount 100
        Maximum metric variance 1

  Automatic Summarization: disabled
  Automatic address summarization:
    Maximum path: 4
  Routing for Networks:
    192.168.30.0
    10.0.23.0/30
    192.168.99.0
  Routing Information Sources:
    Gateway         Distance      Last Update
    10.0.23.1         90           2186597
  Distance: internal 90 external 170
```

Router#

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R1

Physical Config CLI Attributes

IOS Command Line Interface

```

C    192.168.11.0/24 is directly connected, GigabitEthernet0/1
L    192.168.11.1/32 is directly connected, GigabitEthernet0/1
D    192.168.20.0/24 [90/2172416] via 10.0.12.2, 03:03:16, Serial0/0/0
D    192.168.21.0/24 [90/2172416] via 10.0.12.2, 03:03:16, Serial0/0/0
D    192.168.30.0/24 [90/2684416] via 10.0.12.2, 03:00:39, Serial0/0/0
D    192.168.31.0/24 [90/2684416] via 10.0.12.2, 03:00:39, Serial0/0/0
D    192.168.99.0/24 [90/2684416] via 10.0.12.2, 00:04:27, Serial0/0/0
    203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C    203.0.113.0/30 is directly connected, Serial0/0/1
L    203.0.113.2/32 is directly connected, Serial0/0/1
S*   0.0.0.0/0 [1/0] via 203.0.113.1
    
```

```

Router#
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#no ip route 192.168.99.0 255.255.255.0 10.0.12.2
%No matching route to delete
Router(config)#ip route 192.168.99.0 255.255.255.0 10.0.12.3
Router(config)#end
    
```

```

Router#
%SYS-5-CONFIG_I: Configured from console by console
    
```

```

Router#show ip route 192.168.99.0
Routing entry for 192.168.99.0/24
Known via "static", distance 1, metric 0
  Redistributing via eigrp 100
  Routing Descriptor Blocks:
    * 10.0.12.3
      Route metric is 0, traffic share count is 1
    
```

```

Router#
Router#
    
```

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MISSION 7 

1 Final Verification – Check the Network

Q: Make sure the faults from Mission 7 are fixed and the network is working properly. Run the following commands on each router and check the results.

```
show ip interface brief
show ip eigrp neighbors
show ip route
show ip protocols
```

A: The results show all interfaces are up/up, EIGRP neighbors are established, routing tables contain the correct routes, and EIGRP is running (AS 100) on all routers.

Verify the following (as per the addressing plan):

Router	Interface	IP Address	Status
R1	G0/0	192.168.10.1	up/up
	G0/1	192.168.11.1	up/up
	S0/0/0	10.0.12.1	up/up
	S0/0/1	203.0.113.2	up/up
R2	G0/0	192.168.20.1	up/up
	G0/1	192.168.21.1	up/up
	S0/0/0	10.0.12.2	up/up
	S0/0/1	10.0.23.1	up/up
R3	G0/0	192.168.30.1	up/up
	G0/1	192.168.31.1	up/up
	S0/0/0	10.0.23.2	up/up

Also verify EIGRP neighbors and routing tables on all routers.

2 Explain the Packet Journey

Q: How does a packet travel from 192.168.10.10 to 192.168.30.10?

- A:**
1. PC 192.168.10.10 sees 192.168.30.10 is on a different network, so it sends the packet to its default gateway (R1 – 192.168.10.1).
 2. R1 looks up its routing table and finds 192.168.30.0/24 via EIGRP, next hop 10.0.12.2 (R2).
 3. R1 forwards the packet to R2 through 10.0.12.0/30.
 4. R2 checks its routing table and sees 192.168.30.0/24 via R3 (next hop 10.0.23.2).
 5. R2 forwards the packet to R3.
 6. R3 sees 192.168.30.0/24 is directly connected to its G0/0 interface.
 7. R3 delivers the packet to 192.168.30.10.

3 What happens if the EIGRP adjacency between R2 and R3 fails?

A: R2 will lose the EIGRP-learned routes that depend on R3. The routes to 192.168.30.0/24 and 192.168.31.0/24 will be removed (if no alternate route exists). Communication to the Training Center networks will fail from other networks.

6 Final Submission Checklist

Your final report should include:

- 1. Completed Packet Tracer topology (.pkt file)
- 2. Router configurations (R1, R2, R3)
- 3. Verification evidence (screenshots of show commands)
- 4. Mission 7 troubleshooting report (three faults + how they were identified/fixed)
- 5. Packet journey explanation (192.168.10.10 → R1 → R2 → R3 → 192.168.30.10)
- 6. Answers to all 9 exploration questions

4. Exploration Questions

Answer the following 9 questions:

- 1 What happens if EIGRP is removed from R2?**
A: R2 stops exchanging EIGRP routing information with its neighbors. EIGRP-learned routes are removed, so networks that depend on those routes may become unreachable.
- 2 What happens to the routing table when an EIGRP neighbor disappears?**
A: Routes learned through that neighbor can be removed from the routing table if there is no alternate valid route.
- 3 Why does EIGRP not need to be configured on the ISP?**
A: The ISP is outside the internal EIGRP routing domain. R1 reaches the ISP using a static default route, and the internal routers can learn a default path through EIGRP redistribution. The ISP does not participate in EIGRP.
- 4 Why is a default route useful?**
A: It provides a path for destinations for which the router has no more specific route. In this topology, R1's default route points toward the ISP.
- 5 What is the difference between connected, static and EIGRP routes?**
A: Connected (C) – directly attached network.
Static (S) – manually configured route.
EIGRP (D) – dynamically learned through EIGRP.
- 6 What happens if R1 has a static route to 192.168.30.0/24 via 10.0.12.2?**
A: R1 can use R2 as the next hop to reach the Training network. This would make it a static route instead of the EIGRP-learned route (if the static route is valid and preferred).
- 7 Why can a router use a next-hop address that is not on the destination network?**
A: The next hop is the neighboring router to which the packet should be forwarded, not the final destination. The neighboring router then performs another routing-table lookup.
- 8 What happens when a static route and an EIGRP route exist for the same destination?**
A: The router compares their Administrative Distance values. The route with the lower AD is preferred.
- 9 Which route is preferred, and why?**
A: A normal static route has an AD of 1, while an internal EIGRP route has an AD of 90. Therefore, the static route is preferred when both routes exist for the same destination.

5. Final Connectivity Tests

From the Main Campus PC, test the following:

```
ping 192.168.20.10 (✓ Success)
ping 192.168.30.10 (✓ Success)
ping 192.168.20.1 (✓ Success)
ping 192.168.30.1 (✓ Success)
```

Also, from a PC or router, test the ISP path:

```
ping 203.0.113.1 (✓ Success)
```

On R1, verify the static default route:

```
show ip route (look for S* 0.0.0.0/0 via 203.0.113.1)
```